

ActInf GuestStream 109.1 ~ Hemming: "Free Energy, Attentional Narrowing"

GuestStream | Hemming | 1:21:32

Hello, welcome everyone.

Hi.

It is...

Greetings.

It's May 23rd, 2025, and we're in Active Inference Guest Stream 109.1 with Alexander Hemming presenting some shared work on emotional attention and predictive processing.

So, Alexander, thank you for joining.

Looking forward to your presentation, and anyone watching live, please feel free to write questions in the live chat during.

So, thank you, Alex.

Okay, sweet.

So, emotional attention and predictive processing.

For the actual presentation, I'll go through an introduction, a foundations of attentional narrowing, then a understanding of the neurology and a connection between the attentional narrowing and the predictive processing.

I'll run through some computational models, so it'll give a more holistic understanding of this, and then we'll go through a discussion, which will include the applications and future directions.

So, this is the team.

At the moment, I've got me.

There's a co-author for a book I'm writing as well called Dylan, and then there's Kirsten, who is a psychology researcher at the University of Birmingham.

So, this is a project that we're working on.

I'm going to briefly talk about this.

This is a book that me and Dylan are working on.

We're using predictive processing and active inference to understand the basis for how ideologies are formed.

Kind of from a foundational way, building up to pretty much all the differences in politics, but it's difficult to go into at this point.

If anyone has any questions about this, happy to answer as well.

So, introducing the idea.

Here, we have the whole idea, which is, in a sense, when you have a prediction and the prediction error goes wrong about something which is very emotional, then you tend to have an emotional salience of attentional narrowing.

So, the threat uncertainty creates a predictive error minimization process, but that also creates

a neuromodulation of a temperature parameter, which then creates a narrowing effect so you consider fewer alternatives.

The thesis aims are to kind of create more of a unified understanding of the brain by adding together more of an understanding of what the actual parameters in the active inference equations link to.

It's connecting it between a lot of different disciplines to try and understand more of how this works.

Affect-based attention is one of the most prime parts of this.

It's how does emotionally salient information go through a Bayesian distribution to try and make a different precision weighting compared to more calculated predictions, which is the normal direction of predictive processing.

The research gap is predominantly about the lack of a link between processes such as dominant response facilitation and attentional narrowing and active inference.

And so using those two together, it aims at addressing things which have not really been talked about, but also connecting a lot of this together.

I'm running through quite a lot because we have quite a few slides in the time, but the foundation.

So it starts originally in the 1908 theory of the Juerk-Stoldsen inverse U-graph, which many people might know from A-level psychology.

Well, that's a UK thing, but yeah.

And the idea is that if you have a certain amount of pressure, then your most reinforced responses will be facilitated.

So let's say you're playing piano in front of an audience and you've been practicing for a very long time.

You'll actually improve your performance of playing the piano until the stress gets so high that you choke and you do even worse than you did before.

This went on to be understood by Easterbrook as not just all arousal creates that.

It's not excitement which creates this attentional narrowing process, but it's actually a negative effect for it.

So it's the uncertainty of some form of threat will then create a form of pressure.

This form of pressure then facilitates a lack of consideration to alternatives.

It's kind of automatic when it goes into the dominant response facilitation, but he also connects it to belief formation.

It's almost like a societal stress that facilitates a lack of consideration of alternatives when you're creating an ideology or a belief about certain concepts.

I've included a few references in the slides if anyone is interested.

There's quite a few more references, but some of them are a bit key.

So why hierarchical predictive processing?

Well, this is more about the idea that it kind of like connects these things in a way where we can actually add to more of a predictive understanding of the brain as a whole.

It allows for a significant magnitude between interaction between the social phenomena and the predictive mechanisms.

So anything which is dependent on motivational mechanisms has more of an ingrained understanding of it.

So that's the attentional narrowing.

And now we've got the neuroscience behind it, which is one thing which is very strongly researched is the neural gain and emotions as computations perspective.

We do know that having adrenaline in your system, it's a kind of a neuromodulator for these sorts of processes.

And it does create this what I'll show later to be the sigmoid function where the most reinforced, the most reinforced processes are strengthened and the other ones which are more.

So you have a lower probability of transitioning into these states is reduced.

So this is also enhanced by prefrontal cortical inhibition.

So we do know that cortisol and catecholamines do increase the reactivity to the reflexive type of actions.

And they reduce the contemplative, reflexive thought processes.

This means that when you are under a lot of stress, it's difficult to contemplate and think clearly.

But it's quite easy to react to some situations and do the most learned and most reinforced behaviors.

But it's more of an effective neuroscience look at things because it's opening and closing different pathways rather than something which is more ingrained in the computational neuroscience of seeing how excitable different neurons are, such as which would happen in the neural game.

So it's instead of resistance to stochastic resonance and noise, which affects each individual neuron, it would be more of an opening and closing of different pathways.

But it still has the same effects of the attentional narrowing mechanism.

And as we can see from reflexive, it's more about the individual contemplative strategies of the individual.

And as we can see from the reflexive, it's more about the reaction process.

It's more of a slow acting process when we have kind of adrenaline rushing into our system and it affects the entire brain.

So this is different to direct precision racing, which you would normally see inside policy selection mechanisms, because you have this idea that there's a different exploration exploitation dynamic in which you kind of focus on the things which are the most expected to happen.

They become more salient.

And when something unexpected happens about these most expected situations, then you kind of focus on that and then you dive into that type of information.

But that's such a specific dynamical process that it's not something you consider as universally affecting all areas of the brain in a very emergent way where you have an individual neuron and the traits with those neurons actually affecting the dynamics of how the whole brain works.

It's more about the actual processes of how free energy works in the brain.

So stress from prediction error.

Emotional valence is the way in which you would kind of understand stress to come about.

This is if you have a negative emotional valence, then you would have an emotion which does not.

It's not pleasant.

It's something which is kind of like fear, anger, sadness.

These are all negative emotions.

And because of the least action principle, people go down the idea of you want to avoid the negatives and you want to reduce the amount of energy that reduces the amount of free energy.

So all negatives were associated with a rate of change increase with free energy.

But the issue with that was when people use the idea of predictive error minimization, they tend to use the idea of everyone is always predicting everything and everything has to be a direct calculation.

And that has been an issue.

The kind of like rationalizations people go about that is like if you're surprised by a present which a friend has given you, it has filled the prediction of a social expectation of you being given a present.

It doesn't necessarily have to be that type of rationalization, but you can see how these things don't seem to have much intuitive sense behind them.

And that's that's why when we have a Joffley's points, we delve into the idea of like when there's a prediction and it hasn't you predicted something, but it becomes worse to you than what you originally predicted.

Then that is still something you would say has negative emotional valence.

And that's a very simple way of thinking about us.

But it still works in the way of even though it's a negative to what you actually predicted, it is still an increase in free energy or predictive error mechanisms because you can understand that prediction might not be the best word for this.

And a few people have written articles on this.

One person has decided to determine that it's a instead of Bayesian priors, they say a Bayesian preference.

So it's a Bayesian prior distribution and a Bayesian preference distribution.

If we have the Bayesian prior distribution, we can have the idea that there is a calculated weighting of what you expect to happen based on your kind of like prior opinion of that being that probability.

And it's very calculative.

It's very much on this understanding of what is observed is real and you're trying to get around to what is actually an accurate representation of things.

But if you have certain sensory inputs and they might just be like, I'm hungry.

I need this.

And as if people know about the dark room situation, that doesn't always add up in the way that people understand how motivation works and in the way that prediction is accounting for the hunger.

And instead, you kind of would have to have a preference distribution, a Bayesian preference distribution, which can be understood more instead of as predictions, more as desires and drives, which then weights the predictive mechanisms towards a certain direction.

But this at the lower order functioning would still be would still be kind of a Bayesian prior in of itself because it's having an influence in the probability distribution to try and give you more probability to weighting your behavior in that direction.

So it shouldn't really be understood as a Bayesian prior prediction, but more as a driving force, which increases salience of certain ideas.

And this leads into what my co-author and colleague in the article is keen on describing as a Hume vs Kant dynamic.

And the Kantian understanding of predictive processing is very much on their way of understanding the

generative model.

But it does ignore the fundamental idea that you can understand to a certain degree a relationship between desires and passions directing what you want as an individual.

The fundamental thing is, in a sense, would you do anything if you didn't have a desire to move, a desire to exist, a desire to do everything?

Or are you a method for the universe to know itself and you're just an emergent property which has calculations for the sake of it?

It doesn't make sense without there being a driving force, which does count as the concept of a desire.

And these desires in a Bayesian preference distribution can direct individuals' behaviors and even their beliefs towards things that are important to them.

Attentional narrowing in predictive processing.

So, as attentional narrowing relates to all this, you can see that as you have a couple of Bayesian preferences and Bayesian priors and they're in dissonance with each other,

there's predictive error because of the mismatch between these two, and that can create the negative emotional valence.

This negative emotional valence can then stimulate a neuromodulation of an attentional narrowing process, which then affects the way that your decisions would be made.

They attentional narrowing you so you have less exploration to alternatives, thus making your decisions, especially in societal uncertainty, a bit more rigid, a bit more deterministic, and a bit more going down reinforced certain paths.

And this leads into the exploration exploitation trade-off.

Going further into the mathematics, we can understand that there is a tendency to explore a lot of alternatives or a tendency to exploit the very well-learned and very deterministic paths.

And this is understood differently according to different paths.

But we've got the fundamental variable in predictive processing that relates to constant rigidity and attentional narrowing, and that is the omega value.

So this is a stress-induced sigmoid function.

That means it's a temperature parameter, but an inverse of that, so 1 over the temperature parameter.

What happens is we know it is affecting all processes equally as this.

If there was the same sigmoid function for every neuron, then, you know, if there's the same excitability of every neuron, then that would be termed as the sigmoid function, because it is the same.

The most reactive would be more reactive and more reliable, and the least reactive would be the less reliable, making the pathways a lot more deterministic.

And it shows, for example, in Natchatal, we have the understanding of how relationship uncertainty can create political absolutism.

And this is, you can see how inside a precision-weighting mechanism that shouldn't be the case.

It shouldn't be that you've had something with a relationship, so you're going to become more absolutist in an ideological perspective.

They don't relate to each other, but somehow one of them affects the other,

which can only really be understood outside of precision-weighting in a more domain-agnostic kind of

perspective.

Though I don't think I'm using that term necessarily correctly.

Sorry.

So, inverse temperature.

Just to run through what this is, it's the thing in thermodynamics where the more atoms you are hitting, the higher the temperature.

Thus, the higher a temperature you have, the more attention you are paying to, the less narrowed your attention.

And following that, you can see that there is a sigmoid function would then reduce the temperature, because it would make everything go down a very linear path.

You would only have the well-rehearsed, well-learned things which are going to be exploited, rather than the directions which would be a lot more chaotic.

It's almost as though the chaos can add to, like, in the edge of chaos perspective, we have an optimal functioning where we have the most exploration bordering on chaos, with actually having the largest amount of complexity within a system which still has a function behind it. So, cognitive flexibility as the baseline parameter.

This is the equation that we brought in, which is the idea that the rate of change of free energy times by an ability to handle dissonance minused or added to the initial cognitive rigidity of an individual makes the attentional narrowed cognitive rigidity.

And this parameter then could be used in the state change function.

It reflects cognitive flexibility in a number of different fields, because you can see how a lot of people in certain psychopathological diagnoses can tend to have cognitive inflexibility in a way which would support this equation.

And in the computational models, we have the state transition temperature.

So, this one here is the main equation.

It's obviously not mine, but there's the omega value here, which you can see times by the logarithm of the transition matrices, which is the next stage considering what the current stage is.

And so, when you have the log of all these values, then you're pushing them with the high ones to the high point and the low ones to the low point.

And then you're creating a softmax function to guide these all to add up to one.

And that kind of allows them to be accurately mapped out inside a probability distribution.

And this then makes the transformed matrices afterwards.

Sorry.

And here's to go through the maths.

If anyone wants to pause this on here, it might be a bit better,
but I think I went through the maths just then.

And here we can see when we combine it all together.

So, the probability of the next stage, considering the previous stage,
is equal to the softmax of this,

which is the original equation that we came up with
for determining how the negative valence affects
a new attentional narrating process.

And this attentional narrowed original omega variable
then is multiplied by the logarithm of the transition matrix.

And as you can see, when it is low,
then this is the maximum amount of cognitive flexibility.

This is when you can have a very clear mind
and consider all the possibilities before you,
but you may have choice paralysis.

You may not be very comfortable in a fight-or-flight situation
where the pressure of a situation is making you
go to a sudden, immediate action,
which is going to be the most learned action,
and you want to do it very well.

And when you have it as deterministically as possible,
even simple thoughts might be difficult
because you might need to explore
a lot of options around you,
and even the most deterministic thing
could just be to stand there and be frozen in the headlights.

As such, this affects both belief formation
and dominant response formation
because we can see how this affects
every aspect of life in our interests.

It's in politics, but obviously,
it does reflect in psychopathology
and it does reflect in sports psychology.

This leads to the bidirectional dance
of cognitive rigidity.

So we do have the idea that from the ground upwards,

so from the sensing to the actual thoughts,
we have how the environment affects the brain,
and that would be the stresses inside the environment.
And that's in the bottom-up processes.

So as we are in any societal uncertainty,
it will create these prediction errors,
and when these prediction errors are manifest,
they will then lower the temperature parameter,
which then top-down goes to your actions,
which are to have more entrenched reinforced beliefs,
and it is to have a limited...

You may jump to conclusions more easily,
you may be fixated on certain points,
you have an inability to change your mind
about certain things,
and you would also be more likely
to do your well-practiced responses
rather than exploring alternatives.

This leads to discussion.

Right, so...

Flexible minds nuance their ideas.

This is something I find quite interesting
in this concept.

So if we have this inability to transition
from one state to another state,
then we don't really have the ability
to consider other people's perspectives
to add to and build on our ideas.

If we entrench ourselves in our ideas,
if we have rigid minds,
if we have absolutist ideologies,
if we have absolute things inside any perspective,
then we are unable to create
the most complicated,
the most detailed concepts,
which, you know,
in this leading to this ultimate complexity
at the edge of chaos,

it would never be achievable
if we stick ourselves in what would be a...
which I should have addressed
as a suboptimal local minima,
in a sense of...

If we imagine a large landscape of ideas
and we can't explore other places
and we can't get out of this hole of an idea we're in,
then we will never see a bigger hole of an idea,
which is a better hole.

We would never have to get to the globally best hole.

We would always be stuck in the rigid beliefs,
which can potentially lead to a lot of problems.

Whereas having flexible,
nuanced beliefs means
we will have more integrated complexity
in our ideas.

In political theory,
that is how you get to the best conclusions.

It's not necessarily that there's one idea
which is better than another idea.

It's just if you get an absolutist idea,
you tend to have problematic stances
on a multitude of different things.

It's better to consider
as many perspectives as possible
so you're aware of good ideas.

But I believe there's a famous quote of,
it's good to have an open mind
but not so open your brain falls out.

But there are a lot of nuances
to every aspect and every rule
that you can make both for yourself
and for everything in society.

And this is in conjunction
with cognitive dissonance
and predictive processing.
You can have an integration

more thoroughly with this
based off the attentional narrowing details.
So as we have Bayesian-Prize,
something important to you is criticized,
which is in the Bayesian preference distribution
and that in conjunction
with Bayesian-Prize distribution
of calculations
can then create the resolutions
in cognitive dissonance.
And these resolutions are either
affects, cognitions or behaviors.
So these are the ideas
in cognitive dissonance
which are in contradiction
with each other,
which cause the dissonance,
which causes the rationalizations
which make the theory
of cognitive dissonance.
And so if you have an emotional point,
it could be an addiction to cigarettes.
And then you have a fact
that cigarettes are bad for you.
This dissonance between them
creates a prediction error.
And this prediction error
then leads you to rationalize
that cigarettes will help you lose weight
or cigarettes will do...
These are common cognitive dissonance scenarios
which I've studied quite well.
But that establishes how
when you have attentional narrowing,
your consideration of alternatives
is reduced to such a degree
that you're more likely to rationalize
because it lowers your ability

to see alternatives
as you move towards the rationalization,
making the rationalizations more distinct
when things are very emotionally significant to you.
And this leads connected to heuristic adherence.
So heuristic adherence
is very strongly connected
to cognitive dissonance to a degree
and the predictive processing mechanisms
because heuristics,
if you read the Daniel Kahneman thing,
they're generally held in negative light,
but heuristics are also often held inside
just a tool for how you can do things throughout life.
But if you are intentionally narrowed,
then your heuristic adherence is a lot stronger
because to get out of your entrenched beliefs,
you have to rely on something
which is kind of an easy go-to card.
And this can mean that you are unlikely
to nuance these heuristics.
If you don't nuance your heuristics, however,
then you tend to justify perspectives in society,
which to a lot of people can seem very clearly wrong.
But if you're under a very large amount
of emotional strain,
then it can seem more appealing to you.
An obvious heuristic,
which was studied in marketing,
is the price tag.
You rely on the price tag,
but if you're really stressed,
you don't really know what to think,
and you're kind of faced with this time pressure,
then the price tag itself
will be a reliance you'll have
on the actual value of the item you are buying.
But then you also have

one of my favorites of heuristics,
well, one of the worst heuristics,
but that I like to talk about
is the just world fallacy slash hypothesis.
And this is generally defined as
everybody gets what they deserve
and deserves what they get.
In this perspective,
we have a justification for the status quo,
which could be anything
inside the most authoritarian societies.
It can be very distinct cultural dynamics,
which you feel you can't see why they happen,
but you're going to justify them
because they exist for a reason,
because they exist.
And this can cause a lot of victim blaming.
It can cause the belief in a pretender fallacy.
If you're in a situation
where you do not feel like you deserve this
and you feel like something
is not right about things,
it can be due to a backfire with this heuristic.
It can be in the general resistance to change,
which is endemic with the loss aversion tendency
because people are generally against change
even when it's beneficial.
And a just world hypothesis attaches onto that
because the heuristic adherence to the idea
that everything must make sense
or it exists for a reason
makes you unlikely to change,
especially when there is a lot of stress.
So the attentional narrowing mechanism
facilitates the predictive processing
area minimization,
which causes more of a reliance on the heuristics,
which then reduce your ability to change

in these situations.

And this is very strongly studied
in relation to cognitive rigidity
through how cognitive rigidity
is a fundamental neuroscientific trait.
It's more recent in the development
of political psychology,
but it shows that cognitive rigidity
helps or forces people
to be unable to change their mind
and to create what they term
as absolutist ideologies.

So does this better our understanding of the brain?

I think combining attentional narrowing
with cognitive rigidity
and putting them into a computational mechanism
within the active inference models
of state transitions
does help give an understanding of the brain
which enhances the predictive processing mechanisms.
and it should help enhance
every step of its involvement
with the involvement in psychopathology,
in precision psychiatry,
because if you go into the area of psychology,
the importance of how stress affects people
is incredibly significant
and showing if there's biological components to this,
whether they can be reinforced or not,
some innate component,
which is outside of the direct environment,
which makes a difference between people
is also interesting for the diagnostic criteria
of what people go through.

And that's where we get into the applications
of precision psychiatry.

So schizophrenics and autistics
tend to have more cognitive rigidity.

They tend to be cognitively inflexible
is what they term it,
but the resistance to changing their beliefs
and the correlation they've had
has actually led people
to even ask the question of
is cognitive flexibility the missing link?
Is it the component in the differences between people
which cause the genetic component?
Because the genetic markers for autism
and schizophrenia are vaguely strong.
And so the link between these two
is seen to be on the cognitive rigidity,
which seems to be strong in both categories.
Inside predictive processing,
people have also had the idea
of the generative model
and the recognition model
being quite significant
in how the generative model
and the over-the-top focus on predictions
and imagination of the scenario
creates the preference for schizophrenic people
to have hallucinations and delusions
and come up with ideas outside of
what people would normally expect
to be realistic.
And in the autistic situation,
there's the sensory overload,
which is more in the sensory aspects
of the recognition model.
And that's over-focus on the sensory information.
It seems to be more in...
There seems to be a very large link
between autism and schizophrenia
according to the dysregulation perspectives
of active inference.
And that is why there's a relevance

to schizophrenia,
because the idea of the entrenching in beliefs
is strong with the concept of delusions.
So schizophrenics are also more likely
to have conspiracy theories.
And this entrenched belief system
that they hold means
that the cognitive rigidity
appears to be a strong factor
in what develops the problematic issues
with the diagnosis,
with suffering with schizophrenia.
There is also educational benefits.
So there are studies into the idea
of just attentional narrowing in general,
increasing your focus.
And if you have a certain amount of pressure,
then it does help you study harder.
This is important in a lot of developments
in psychology,
which can use the concepts
of attentional narrowing,
especially in a mechanistic way,
to develop into other psychological concepts,
which then help understand
how a drive towards supporting
a value and ideal you believe in
can be facilitated
by the pressure of this ideal.
It is almost as though
when you have a bit of stress
in your life,
sometimes laser focusing on a task
can reduce that stress.
Doing something which is associated
with your values can reduce
an error process.
It can do an error minimization mechanism

for the actual issues at hand.
And this also can understand
how learned helplessness
can happen from an error minimization process
where you see what you are doing
as being difficult to solve.

So instead of when the attentional narrowing
gets too high,
you will go down to a simpler solution,
which is learned helplessness
or giving up
or disassociating from the concept.

Instead of focusing on the task,
the anxiety will get too high
and it'll be harder to focus on that task.

And understanding that inside
a relationship between other metrics
inside the brain
can help understand
what can cause learned helplessness
and what influences can develop that.

Further, a respect for academia
in total can facilitate
the importance weighting
and the Bayesian prior distributions
in favor of doing better in academia.

And it can actually,
according to certain texts,
for example,
the book The Moral Arc,
there's evidence that
respect for academia
and having a skeptic-first mindset
can facilitate a sense of morality
in any country
which holds those values,
which I think is quite significant.

And future research directions

after the applications.

This is in social and political psychology.

These are areas in which I think

attentional narrowing

and cognitive rigidity are profound.

Obviously, cognitive rigidity

is pretty good for cognitive rigidity.

If we have a computational aspect to that

and we attach it to the mechanisms

of predictive processing,

then it will distinctly help.

The right-wing authoritarian personality trait,

which is a development

from something which is called

the fascist scale,

which applies to both

the far left inside

a traditionally left country

and the far right

in a traditionally right-wing country

most of all.

but it's still called

right-wing authoritarianism

because of the associated

of right-wing

with traditional,

conventional attitudes

and that being correlated

with the concept

of authoritarianism.

Social dominance orientation

is kind of the idea

of feeling as though

there should be a dominance

over certain groups

and all of these seem to associate

with the concept

of heuristic adherence

in the political environment.

It's the adherence to rules,
the idea that everyone gets
what they deserve,
deserves what they get
because of this authoritarian structure
that you live in.

That enhances by the intentional
narrowing focus
and as such,
you can kind of see
how they would not nuance
their beliefs
according to these dynamics
of their society
and they would go
more absolutist
into supporting
of any ideological background
especially when it's
strongly associated
with a Bayesian preference.

And that works
for all of these processes
and they're all
psychology theories
which are very interesting.

And in social psychology
you can see
how they also relate
in the same way.

Game theory

I'm not the most certain on
but social facilitation
is the basic one
about dominant response facilitation.

Conformity and distress
obviously would increase

because you have the uncertainty
enhancing the attentional narrowing
even though it's not
directly precision weighted.
There's the threat uncertainty
significantly influences
your chance to conform.
And even in the Dunning-Kruger effect
there's a preference
to narrowing your focus
to only what you've experienced
and not to an exploration
of unknown possibilities.
And so this process
is believing you're better
than you are
because you don't have
much experience
about what else is out there
which probably happens
in the area of research
and I'm hoping
it doesn't happen to me too much
but I think that
there's the idea
of exploring other options
instead of going down
your singular path
which gives you more
an awareness
of the unknown unknowns.
and then there's reactive
approach motivation.
So this is something
which I think is a really good
direction to take future research
after attentional narrowing
because this gives more

of a holistic picture
of how when under stress
you will do certain things
which then make you more zealous.

I have a quote here
which is
if any adaptive change
in the brain
will minimize free energy
then how does
an adaption to stress
minimize free energy?

And the adaption to stress
doesn't always minimize
free energy
in that regard.

It aims to minimize free energy
but because of the positive feedback loop
it can loop things
through quite a lot.

and in the way
that reactive approach
and motivation
tries to react to stress
reactive approach motivation
is that they find
that any societal stresses
will be reduced
by disassociation
from these societal stresses
through a laser focus
in an ideal
that you believe in.

You can distract yourself
by focusing on something
which to you
has value and worth
and that does increase

for example
zealous adherence
to a cult-like religion
in opposition
to when you are
suffering quite a lot
in a lot of threat
and uncertainty
in your personal life
and that can be enhanced
by cognitive rigidity
when maybe the stresses
are more effective.

And then this is
the conclusion
summary of points.

So from this
we've got
attentional narrowing
from emotional
stresses.

This creates
the predictive error
minimization
from emotional
stresses

creates an increase
of rate of change
of free energy

which then
forces you

to rely

on heuristics

when your attention
is narrowed

and you are too rigid

inside your perspectives

because you then

don't nuance
these heuristics
and then you
have a perspective
on both society
and actions
in society
as well as
a facilitation
of more learnt
like practiced
piano responses
in front of an audience
type of situations
of the most reinforced
neural connections
and belief formations
and belief
that you don't really
have belief updating
sorry
and it develops
a more authoritarian
outlook on
things in society
and thanks
and if you have
any questions
then you can email me
and also you can ask
for the email
of either of my
colleagues
well mostly Dylan
and Mr. Grove
and it would
be much appreciated
thank you Alexander

so Daniel
yes
awesome
thank you
while I'm
cropping things back
maybe share a little bit
like how did you
come to this project
or how did you
come together
with these authors
and what are you
like looking for
in terms of milestones
in that collaboration
well
I've
been the co-author
with Dylan
for a while
and we've been
writing a book
which is the
foundations of ideology
so our
our work
in trying to
you know
explain the
political dynamics
we came to
quite a lot of
conclusions about things
when we did research
very continuously
coming up with a lot
of ideas

and in our research
we've read quite a few
books on
between us
on active inference
and this seems to be
a very prominent gap
what I hope to gain
from it
is to add to the
community
and to try and
help as much as
possible
well
to ingrain myself
inside research
and
well
the book has been
my passion project
for a while now
and it's
and it's now
Dylan's
passion project
to a degree
as well
and we do
believe that
this will
be good
to add to
the research
community
and it will
help the world
in the theory

of how
you know
you can
address certain
things which
will help a lot
of people
inside a
diagnosis
and in
political dynamics
reducing the
polarization of the
world for example
okay
a lot of
topics that we
could go into
there's a few
questions in the
live chat
and anyone can
ask more
questions
so I'll read
some of those
maybe leave the
slides up
and we could go
to a slide
if it comes up
okay
alright
Susan asks
is there any
research on
addiction to
exploitation

yes
I mean
addiction to
exploitation
well the
exploitation
exploration
trade-off
is more of a
machine learning
term
but the
thing about
addiction
is actually
more towards
the policy
selection
process
and that's
not the
state transition
process
necessarily
and that
addiction
mechanism
is more
about
kind of
the gamma
temperature
variable
rather than
the omega
temperature
variable
and it's

kind of
more about
the
dopamine
pathways
and how
you are
reinforced
into
having
I'm not
super
knowledgeable
about that
area
but I know
it's more
associated
with dopamine
it's more
associated
with policy
selection
rather than
state
transition
and there
is a lot
of research
on that
from what
I'm aware
okay
here's a
question
that I
wrote down
well first

the scope
is very
vast
essentially
encompassing
our human
cognitive
niche
ranging
from
differences
in
neurodiversity
associated
with like
context
changes
and different
scenarios
in a given
day-to-day
life
to just
different
people
different
stages
of life
and then
all the way
into these
social pieces
so
how
do we
think about
these more
system

general
cognitive
phenomena
with the
domain
specific
phenomena
and then
also
multi-step
kind of like
rhetorical
or justification
phenomena
like what
do you think
that model
kind of
looks like
the equations
do give some
pointers
and reference
but
what do you
see the
puzzle pieces
coming together
looking like
to bring
together
like
fundamental
constraints
of informational
systems
with these
system

general
stress
and
attention
phenomena
with
ultimately
the kind
of domain
specific
models
that are
usually
deployed
in a
specific
generative
model
well I
mean I
think the
specific
generative
models
can work
the best
as they
are and
then this
model just
basically goes
that when
you explore
in a sense
in any
part of
the Markov
chain

you are also
forcing on
yourself
more of
a rate of
change of
free energy
it's almost
like any
action that is
taken will
theoretically
in their
increase of
free energy
they will
potentially create
stress
it's like
thinking
like if you
were playing
chess
you would
force
yourself
to feel
stressed
just to
try and
solve a
puzzle
and that
forcing
yourself
to feel
stressed
means you

are creating
a tiny bit
of stress
which then
narrows your
thought processes
so it's not
that it's that
this stuff
happens anyway
but there's
very constraining
mechanisms in
place which
just stop
too much
free energy
accumulating
during that
same
domain
specific
process
if the
domain
specific
processes
are
specific to
going along
one direction
from one
step to
another
in a
Markov
chain
rather than

in a
one thing
affects
everything
dynamic
thanks
yeah
that was
one of the
parts I
found most
interesting
in sort
of the
deep
affective
work
and the
way that
you brought
it out
and worked
it here
is like
the re-entry
of the rate
of change
of free
energy
into an
equation
because otherwise
the
the free
energy
values
or the
rates of

change
can be
essentially
epiphenomenal
like a
model diagnostic
used in
training
but then
that generative
model
doesn't have
re-entry
or causal
efficacy
of the
free energy
itself
but then
it is this
really
clever way
to bring
global
dynamics
back into
the process
of the
domain
like
staying with
that chess
example
there's a
time pressure
and time
pressure
and choosing

when to
act
is not
always
modeled
in
decision
making
models
it'll
just be
this kind
of like
frozen
universe
and then
a slot
machine
is pulled
or a
frozen
universe
and then
a move
is made
in a
teammates
so that
when you
start to
bring in
the
unfolding
dynamics
it's
this kind
of like
everything

plays into
that free
energy value
or rate
of change
and that
provides
the domain
agnostic
signal
of
basically
how well
things are
going
and then
valence
has been
associated
with whether
that is
going
better or
worse than
itself was
expected to
be
yeah
yeah
perfect
I like
the analogy
okay
another
question
from
Susan
could

cognitive
rigidity
be related
to a
deficit
in
perceptual
affordance
acuity
I think
perceptual
processes
are a
different
neuromodulator
in the
way in
which I
got the
state
transition
from
a
2024
article
there was
a
acetyl
choline
parameter
for
perception
specifically
I'm not
too keen
on the
knowledge
of

perception
but I
do know
that there
is
in that
area
something
more
it's a
different
neuromodulator
that's
as far
as I
can go
I do
think
when you
look at
neurotransmitters
and hormones
inside the
brain
there is
a
tendency
for
neurotransmitters
to slightly
affect
other
pathways
if you
look at
how
stochastic
resonance

which is
the
noise
between
neurotransmitters
affect each
other
it's not
necessarily
that you
have to
have the
thing which
is normally
activated by
that neurotransmitter
is only
activated by
that there's
like a slight
effect from
others
so I
think
adrenaline
could affect
your vision
as well
but I
think it's
more about
the
ability to
change your
states inside
those processes
though I
don't know

too much
about that
area
okay on
I think
it was
slide
30
is that
30
oh wait
no I'm
getting the
wrong way
cool
okay
yes
yes
all right
so
maximum
cognitive
flexibility
is being
associated with
a uniform
probability
distribution
over
future
states
so
um
yeah
that's
that's not
true
really

yeah
how do you
temperature
sorry
well it
can't be a
uniform
obviously
it's
almost
just that
there's a
temperature
parameter
adding in
which is
making you
more rigid
and only
more rigid
it doesn't
it doesn't
ever go
in the
opposite
direction
I mean
there is
the idea
that if
you are
happier
about things
then you
are actually
more
cognitively
flexible

and there
are studies
which show
that
but I
don't think
it can
ever get
to a
uniform
perspective
that's
just a
theoretical
perspective
of you
know
if you
have
temperature
and you
put it
in different
directions
if I could
see that
in anywhere
of an
actual
possibility
I would
basically
think
there's
someone
stuck
in one
place

unable
to make
any
decision
about
anything
because
they're
trying
to see
everything
with equal
possibility
and it's
kind of like
when you have
the edge
of chaos
concepts
so you
have
the
ordered
structure
overfitting
and then you
have the
over exploration
actually into
chaos
you know
you'll have
strange
attractions
you'll have
other things
but it's
it's

it's
not really
information
it's not
information
in enough
of a structured
way to
mean anything
in a goal
directed way
it is just
chaos
and anyone
who is actually
pushed to that
point of exploration
they would not
be able to have
any structured
beliefs and they
would devolve
to chaos
but it would
probably be
closer to
aprasia
it would be
the sense
of like
you know
you wouldn't
have
hallucinations
or anything
because if you
have that much
chaos

a hallucination
is still
a structured
dynamic
process
where you
have
idea
to
idea
idea
it's almost
a hallucination
I would say
is more rigid
because you
can't escape
the hallucination
even if you
want to
whereas
this would be
more you
can't even
have a simple
thought because
you're considering
so many
possibilities
you're completely
changing all
the time
that's the way
I think of
it really
yes
interesting
so

the
more rigidity
is just
crystallizing
or just
sharpening
the definition
on a given
on the given
contours
of a belief
in this
situation
it's a belief
about the
consequences
of action
the rigidity
I feel like
belief is
is a
oh wait
sorry
so the
wait
sorry
what is the
belief
will the
B matrix
are beliefs
about how
things are
apt to change
conditioned upon
what actions
are taken
yeah

well I was
thinking more
in a
distribution of
beliefs
which are
going to
trigger other
beliefs in the
future
so if you
have a
Bayesian
prior
which is
kind of like
if it's
creating an
attractive dynamic
or flow into
a certain
direction
then what's
the chance
of that
happening
considering
those chances
but then if
you multiply
that by
temperature
then you
would potentially
make it more
rigid and not
transition into
the next stage

so sorry
can you repeat
the question
one more time
I'm just
trying to
work out
what is being
made more
rigid
in the
B matrix
and
so
yeah
well it's
it would be
beliefs are
more rigid
in terms of
dominant response
facilitation
it would be
you
are being
made more
rigid in
your
your
it's
the fluid
dynamics in
the brain
of information
so the
rigidity
would be
the attractive

dynamic
it would be
in a sense
the information
flowing
throughout the
brain
is going
through a
singular
path
and because
it's going
through that
singular path
that means
that you're
not going
to
have
more
different
options
that you're
considering
you're
going
to
only
have
the
singular
dynamic
of
playing
the
piano
inside

that
one
direction
but
in
a
belief
it would
be
that
you
would
having
more
rigid
beliefs
it would
be
you have
more
entrenched
beliefs
being
facilitated
more
which
means
that
when
someone
is
trying
to
convince
you
of
an
idea

and
your
beliefs
are
very
entrenched
and
you're
under
attentional
narrowing
then
the chance
that your
belief
will change
will be
very minimal
I think
yeah
I think
I hear
you
and it
may be
interesting
to parse
out
flexibility
as the
capacity
to
change
from
flexibility
being
the
sharpness

of a
given
distribution
in a
moment
because
you
could
have
a
very
sharp
belief
distribution
about
successor
of the
next
states
or whatever
it happens
to be
and that
could be
rapidly
changing
or it
could be
fixed
and then
you could
have
sort of
closer to
a flatter
distribution
that also
could be

changing
but there's
less room
for change
the flatter
you get
with this
extreme
case
like you
brought up
it's like
yeah
does a
blob
have
flexibility
it's like
it doesn't
even have
enough
differentiation
to have
flexibility
because it's
just sort
of
inert
question
from the
live chat
yeah
what did
you mean
by temperature
that's a
great
question

temperature
well it
came from
thermodynamics
so in
in the
understanding
of temperature
in thermodynamics
you have
more
entropy
which is
more information
happening
so
so
when I
mean
temperature
it's
it's
it's
originally
the term
of just
more
stuff
happening
and then
in this
it's used
in the
normal
kind of
reinforcement
learning
equations

of the
amount
of
options
being
considered
so
in this
it is
when you
when you
multiply it
by
the
matrix
which is
the
amount
of
coordinates
in a
graph
which are
pointing
in a
specific
direction
so it's
like a
vector
database
it's
if you
imagine
sorry
it's
if you
if you

imagine
weather
and convection
currents
in weather
they are
pushing air
in one
direction
and if
you're
mapping out
a still
image
of this
weather
you'll have
arrows
moving in
different
directions
from where
the convection
currents
are pushing
things
you can
then give
numbers to
all these
arrows
so all
of these
arrows
are
kind of
like
points

inside
this
graph
and
when
you
have
these
points
in
these
graphs
you can
give
them
numbers
and
sorry
this is
getting
way too
complicated
but
these
numbers
then
represent
the
amount
of
push
inside
a given
direction
and
the
probability
of it

going
in that
direction
and
if you
put this
into a
probability
distribution
like in
quantum
mechanics
it represents
the same
thing
as in
fluid
dynamics
and inside
graphing
temperature
convection
currents
sorry
and
then
the
probability
of
one thing
moving from
one thing
to another
thing
is going
to be
like 10
or 2

and
maybe higher
or lower
but if it's
a high probability
then the
temperature
being lower
means that
sorry
so that's
the point
when you
have
temperature
lower
the
high
probability
is
raised
in
its
number
when you
have
temperature
higher
when you
have
temperature
lower
also
the
lower
probability
is
made

lower
sorry
I'm not
really
in the
best of
zones
for this
but
temperature
adjusts
the
probability
so it
makes
more
probable
actions
more
likely
and
less
probable
actions
less
likely
yes
I'm using
it in
this
yes
I think
about
Kelvin
the
temperature
scale
and how

there's
basically
a stopping
of molecular
motion
and so
if we
were to
have a
ball
rolling
on a
table
and the
table
were to
be
unmoving
and we
were to
drop
the ball
on
like
one way
to say
it
is
it
would
find
the
lowest
point
or a
little
bit
of a

looser
way
is
it
would
reflect
the
actual
distribution
of
the
hills
and the
valleys
and then
as you
shake
it more
and more
with higher
temperature
those
differences
start to
matter
less
in elevation
because there's so much
energy being added in by the
movement of the table
that the small
differences in
elevation are
basically outweighed by the
random
jostling
yeah
yeah

finding the local
minimum
through gradient
descent
yeah
good metaphor
okay
okay
another
question
from
yeah
go ahead
then I'll ask
another
question
oh
sorry
I was just
going to say
yeah
oh
sorry
no
carry on
please
sorry
sorry
go for it
so
well
I would say
I guess
the
only way
to escape
one of those
dips

is to add
a bit
of chaos
into it
and that
chaos
can also
be understood
as a
temperature
because
it's not
really efficient
to try
and get
out of a
correct answer
which is
the dip
to find
a bigger
dip
so you
have to
have something
which is a bit
chaotic
like temperature
yeah
I think
the way
that you
brought in
the u-curve
response
is going to
be really
interesting

to see
as you
develop
this out
because
like that
reflective
and
and
reactive
or
those
on the
previous
slide
stress
may
increase
the
piano
playing
a little
bit
but
then
it
doesn't
have
a
monotonic
relationship
but
then
often
times
when
we're
working

with
a
preference
vector
or
something
like
that
it's
kind
of
like
you
just
increase
the
preference
just
forever
so
it's
actually
hard
to
bring
in
those
u-curve
like
responses
but
those
have
a lot
more
biological
validity
when

we
start
looking
at
the
tension
between
different
general
systems
yeah
that makes
a lot
of sense
if there
was a
preference
for
something
I guess
any
contradiction
to that
would
create
more
free
energy
which
would
then
create
more
attentional
narrowing
in that
regard
okay

I'll
ask a
question
from
the
live
chat
Vitor
wrote
could
trends
be
identified
in
real
time
or
near
real
time
do
you
think
I'm
thinking
of
say
collecting
personal
data
and
establishing
a
personal
baseline
I
think
that's

more of
psychology
research
than
computational
neuroscience
but
you
could
definitely
find
trends
in
real
time
and
you
can
validate
certain
things
based
on
how
likely
someone
is
going
to
like
if
you
weight
the
importance
someone
has
towards

a
certain
thing
then
it
can
definitely
be
collected
in
real
time
based
on
surveys
and
certain
bits
of
information
and
then
also
you
can
probably
do
galvanic
skin
response
things
to
check
how
much
aversive
arousal
or negative

effects
someone
will
have
and
then
see
how
their
attention
will be
narrowed
towards
that
especially
when
considering
criticisms
to
things
which
they
hold
very
important
to
them
which
would
then
be
more
entrenched
beliefs
and more
rationalization
of these
perspectives

it's
possible
but
it's
realistic
and
it
doesn't
really
I
think
it's
not
based
on
this
type
of
research
necessarily
but
it
is
something
which
is
more
based
inside
the
theories
it's
building
towards
in
psychology
already
however

in these
models
you're
talking
about
observables
and
latent
states
of
people
so
conceivably
that
does
connect
with
some
kind
of
self
report
or
biometric
or other
behavioral
measures
that would
be used
to
parameterize
the
model
so
what
would
that
look

like
and
how
do
you
see
that
being
brought
into
practice
well
I
think
it's
possible
but
it
wouldn't
be
super
accurate
the
kind
of
explanatory
power
of
this
process
is
pretty
good
and
you
can
have
some

validity
to
it
you
could
probably
do
some
significance
test
towards
it
but
what
you
would
probably
do
is
just
copy
Lios
Migrod's
research
on
cognitive
rigidity
where
she
finds
that
cognitive
rigidity
through
card
assortment
tests
correlates

with
the
absolutist
ideology
and
that
then you
can
connect
that
to
some
other
people's
research
on
cognitive
rigidity
which
is
associated
with
attentional
narrowing
and
then
if
so
these
things
are
quite
well
established
and
then
using
the

active
inference
equations
in support
of that
just
more
gives
an
explanation
for it
rather than
giving
an
enhance
to the
predictive
power
of it
it
can
the
influence
of
computational
modelling
with
all these
is probably
beneficial
and if
you had
EEG scans
on
people's
heads
you could
probably

see the
processes
of error
minimisation
in conjunction
with this
but
it would
probably
look very
close
to
the
research
by
Lior
and
it
probably
wouldn't
be
self
report
either
it
would
probably
be
just
I think
it's
implicit
cognitive
rigidity
tasks
that
she
does

most
of
the
time
I
don't
think
she
actually
does
much
on
the
self
report
side
of
things
okay
kind of
zooming
out a
bit
how do
you
think
ideology
will
evolve
or
change
as
cognitive
science
brings
new
insights
into

neuromechanistic
aspects
and
real
time
and
different
influences
on
ideology
because
it's
been
I
think
as
your
interdisciplinary
approach
is
showing
there's
like
political
analysis
of
psychological
analysis
of
so
as
these
puzzle
pieces
come
together
what
is

that
going
to
look
like
and
what
ideology
comes
out
of
the
wash
there
yeah
well
I
mean
I
think
there's
this
tendency
for
people
to
always
try
and
point
out
gaps
!
to
try
and
support
a

perspective
that
they'll
have
and
this
filling
all
of
the
gaps
gives
enough
of
a
reasoning
to
try
and
go
this
is
correct
rather
than
this
and
there's
a lot
of
ideological
perspectives
that
people
have
which
are
a bit

problematic
and
if
you
have
so
much
science
refined
around
the
idea
of
it's
bad
to
reject
nuance
in a
sense
then
it
builds
up
evidence
for
that
perspective
I
also
think
that
it
can
help
in
structure
societal

change
and a
predictive
mechanism
for people
inside
debates
with
elites
etc
and
these
things
can
add
to
a
better
society
and
reducing
the
polarization
but
if
we're
talking
about
the
changes
of
ideologies
with
consideration
to
these
things
I

think
it'll
only
be
if
someone
considers
what
we're
saying
then
there's
so much
evidence
behind
it
that
it
can
show
it's
bad
to
be
absolutist
basically
it's
bad
to
not
be
nuanced
inside
any
direction
anyone
takes
you

know
it
encourages
everybody
no matter
who it
is
to
be
a bit
more
nuanced
and
maybe
Socratic
and
question
themselves
and
never
be
fully
certain
about
anything
hmm
how do
you feel
that that
might come
into play
in
power
discussions
like
how would
this
how would

this
for
somebody
let's
just say
experiencing
that
rigidity
how
would
this
land

or
what
if
rigidity
remains
a viable
cognitive
strategy
even in
light of
knowing
about
it
like
in
almost
a
sort
of
cooperate
defect
fashion
like
a global
optimum

could be
reached
with
two
flexible
parties
but one
rigid
and one
flexible
leads to
a different
outcome
yeah
well
I
I think
if we're
talking about
rigid inside
the temperature
dynamic
then it's
good at
going down
a path
but if
there's
the option
of both
exploring
and going
down that
path
then it's
very unlikely
that it's
going to be

correct
it's just
there's a
statistically
less chance
the arguments
of who will
be better
inside any
political
dynamic
that's
not really
necessarily
you know
I can't
say either
way
you know
it's
impossible
to predict
who's going
to be
better
in any
situation
but the
probability
that an
undeviated
path would
be better
is quite
low
because it's
a rejection
of

considering
alternatives
it's not
a
rejection
of something
because it
could be
wrong
even if
you're
considering
alternatives
you could
still go
down the
same
path
but if
you're
just not
considering
alternatives
you're
just
lowering
your chances
of success
and that's
what I meant
more by
the rigid
and if we
have the
assessment
of the
people in
power

then it's
more treating
people
like they
have
like if
they are
overly rigid
and they're
you know
like in
charge of
like King
Jong-un
in charge
of North
Korea
it would
give a
perspective
of analysing
them not
as a
rational
party
but as
someone who
can be
predicted
based off
what they
find is
important to
them
and what
they're
likely to do
on very

narrow rigid
circumstances
it's
you can
probably
give
diagnostic
criteria for
them of
what they're
likely to
consider in
a lot of
ways and
you can
also point
out that
because of
their rigidity
and their
lack of
consideration of
alternatives
they're very
unlikely to
act outside
of very
predictable
metrics
according to
the dynamics
they have

chosen to
act
currently
yeah two
things that

makes me
think of
first the
more
templated
or
rigid
a given
actor is
the more
predictable
their behaviour
is in terms
of their
sense making
and decision
selection
which actually
can be a
strategic detriment
so open
openness
or flexibility
it leaves
the possibility
open
for action
that's difficult
to predict
and then also
when we start
to see the
reasoning type
models
so again to
the sort of
earlier question
of how do

Markovian
generative models
come to
capture
multi-stage
reasoning
like processes
and how do
we model
the duration
of thinking
in behavior
basically this
reflective
to
reflexive
continuum
the more
rigid
focusing just
on a language
model example
the more rigid
the distribution
of the next
token
the less
thinking
matters
because it's
just being
captured in
small attractors
whereas the
larger the
vista
for reasoning
the more

combinatorics
and the more
semantic
exploration
can happen
in that
chain of
thought
space
and so
there can
actually
be
semantically
important
considerations
and kind
of a space
for there to
be
adaptive
updating
so

and then
there's a
fundamental
trade-off
because there's
the time
cost of
thinking
and there's
also
like
rumination
and so on
so there's

no free
lunch
there's no
general
strategy
that can be
you know
in a system
independent
way
and that's
not what
the framework
does
as you
basically
said with
the political
case as well
it gives us
these state
spaces
where these
very grand
trade-offs
can be
elucidated
and they
apply
importantly
to different
scales of
deliberation
like reflective
and reflexive
could be
happening
to a given

person or it
could be a
group
deliberative
process
and so that
points the
way to
very interesting
synthetic
approach
to understanding
the first
principles
of biology
and ourselves
as
political
yeah
I mean
100%
it's
the
idea of
exploration
meaning that
you know
you're in the
exploratory thing
so there's more
exploration
the more
thinking that
happens
and if you're
in a rigid
thing
the more

dramatic
changes
that happen
I think
is quite
apt
but it's
also
interesting
to consider
that if you
are theoretically
open to a
pethora of
different options
and you are
exploring quite
a lot
then you are
also going to
have more of
a normally
distributed
perspective
on things
whenever there
are
whenever there's
a chance
of noise
cancelling out
so that
sounds a bit
odd but
basically
when you
have
a

massive
amount of
exploration
and you're
considering
everything
sometimes
you can
look the
same as
everyone else
who is
considering
everything
even though
you are
theoretically
thinking
more
and you're
doing a
lot
more
you give
the appearance
of being
more predictable
and more
of all these
things
and you
probably are
to a degree
just because
you have
a normally
distributed
consideration

of a lot
of different
alternatives
rather than
going down
one absolute
direction
and then
dramatic change
to another
absolute
direction
so there
are
issues
with that
but yeah
the reason
why I was
more interested
in the predictable
elements
of authoritarians
and their
rigid mentality
is because
you can
through a
nuanced
understanding
of them
give more
of a
distinct
prediction
to them
than you
could

around
someone
who is
exploring
all the
things
all the
time
and trying
to make
accurate
predictions
of everything
through a
non-rigid
mentality
and it's
sort of
the
U-shape
curve
the
entropy
curve
returns
again
where
in one
game
situation
the same
move
is always
selected
in the
other
it's a
uniform

distribution
in both
cases
while it's
easier to
predict
the
one move
player
than the
specific
square
you do
know the
distribution
that the
uniform
player
is going
to select
so it's
it has
a
predictability
but not
in detail
like the
one move
player does
and then in
the middle
are sort
of strategies
that have
all this
different
complexity
in terms

of actually
having
adaptive
possibility
so what
are your
next
moves
or how
could people
provide
feedback
or what
are you
looking to
kind of
get to
next
what are
we looking
to get to
next
well
integrate
ourselves
with the
active
instruments
community
more
talk about
as much
development
as we
can
inside
the
area

we're
very
interested
in the
writing
of this
book
that we
have
and trying
to get
a career
in the
computational
neuroscience
of politics
which would
be pretty
interesting
but
well
we're
exploring
all our
options
and trying
to help
out people
along the
way
I do
like the
idea
of the
sophistication
in game
theory
as the

predictable
nature
of things
but I
think
there was
an interesting
study on
artificial
intelligence
and general
intelligence
and the
deep learning
of AI
and it
kind of
showed
that when
they trained
a chess
bot to
predict one
move ahead
it was actually
better than
when they
changed a
chess bot to
predict two
moves ahead
except through
five times the
amount of
data
so there's a
strange thing
which is that

a higher
level
sophistication
is more
like you're
just trying to
go into
the chaos
of the
world
whereas if
you try and
do everything
one step
at a time
a lot of
the time
it actually
adds out
to something
better
well
there's a
joke we've
had for some
years like
active inference
only works
for one
time step
but you
get to do
it again
and again
and you
can load
everything
into that

time step
but you
only get
one
because if
you were
able to
plan out
trajectories
of multiple
it's
equivalent to
saying that
you had
already
preimagined
all of
what you
could have
received
and if
you want
to do
a rollout
with all
that you
could have
received
from your
first action
in terms
of all the
ways that
that could
influence
your second
action
even with

a small
state space
you're already
getting into
these absurd
rollouts
so that
kind of
is the
justification
for having
action
perception
loop
that's
ongoing
whether it's
called
runtime
or real
time
or online
inference
or just
active
inference
even when
people don't
really come
at it from
the free
energy principle
side
they're just
thinking about
online learning
and about
systems that do

active data
sampling
those kinds of
processes
processes
it kind of
reaches to
the same
place
which is
like
there is
information
in
the
consequence
of
action
such that
you don't
know what
you're going
to do
by taking
two steps
ahead
like you
would by
going one
and then
one
and that's
operationally
how it
proceeds
either way
and that's
the sort

of paradox
of planning
where planning
is commonly
modeled
going back
to again
same with
our chess
metaphor
they're modeled
as these
trees
of different
possibilities
and branching
but we know
that that is
not what
happens
in the way
that planning
proceeds
or even
if planning
explicitly
proceeds
in analog
systems
in the same
way
yet it can
be approximated
with this sort
of branching
counterfactual
play
however

in the one
time step
limit
is sort
of where
biological
systems
have tuned
themselves
to
yeah
makes sense
I think
it's easier
to think
of things
in
exploitative
terms
because there's
less free
energy
so it's
I think
maybe people
delude
themselves
to thinking
that they
can
climb
three steps
ahead
just because
I don't
want to
think about
the other

steps
anyway
I want
to think
everything's
this easy
but I
don't know
yeah
well in
a
reward
or pragmatic
value only
setting
if you're
considering
one time
step at
a time
you do
get
greedy
behavior
but then
if you have
epistemic value
as a real
contender
then
you can
still have
action
selection
reflecting
learning
in that
next time

step
because
the
expected
information
gain
is
legitimately
valued
and so
that can
even
say
well
I know
there's
going to
be a
slight
benefit
if I
go down
that
gradient
but there
might be
other
information
if I
go up
the
gradients
of
pragmatic
value
and so
by bringing
the information

gain
into that
next time
step
then at
least there's
the possibility
of making
co-accounting
for epistemic
and pragmatic
value
then that's
exactly what
the kinds
of modifications
and attention
variations
that you're
describing
get to
I think
so
I think
there's an
interesting
study
which is
the
study
of
tit for
tat
and it
kind of
showed
that
consideration

of
if you
try and
do the
prisoner's
dilemma
over and
over again
the idea
in a
study
where
everyone
says
the best
things
you can
do
for
yourself
is to
focus
on
only
doing
the
thing
that
will
betray
the
other
person
and
selfishly
get you
the best
points

and you
should
always
choose
that
option
inside
this
prisoner's
dilemma
but in
the study
where they
did it
50 times
in a row
and it
was
computer
programmings
trying to
do them
against
each
other
everyone
thought
let's
make
the
selfish
bot
which
can
try
and
take
advantage

of
the
other
people
because
we're
competing
to try
and
who
has
the
best
bot
and
all
the
selfish
bots
did
a lot
worse
it was
the
most
forgiving
bots
did
the
best
it
was
people
who
did
it
better
than

a
tit
for
tap
basis
across
all
the
examples
of
interactions
they
went
into
they
did
the
best
and
it's
it's
it's
not
that
in
each
individual
situation
they
did
the
best
because
they
would
have
a
bot

which
could
either
does
show
your
thing
about
doing
things
in
an
individual
basis
you're
doing
it
one
step
at
a
time
then
that
would
be
a
lot
worse
for you
as
an
individual
but
if
you
have
consideration

to
this
vaguely
nuanced
perspective
of being
cooperative
in
society
being
a
cooperative
organization
or a
cooperative
country
across
all the
interactions
you
have
even
if
you're
doing
it
in
I
draw
or
I
lose
because
of
the
plethora
you
do

a
lot
better
than
someone
who
does
things
selfishly
and
that's
I think
very
similar
to
that
totally
agree
awesome
yeah
it's also
if you're
interested in
global
change
game
it's
this
thing
when
they
put
people
who
have
very
strong
right

wing
authoritarianism
into a
game
and it's
a
four
hour
simulation
of
controlling
the
whole
world
and
when
they
put
very
strong
super
authoritarian
people
inside
the
game
all
together
they
started
off
giving
each
other
wedgies
wrote
on
a

piece
of
paper
look
after
your
own
take
care
of
your
own
type
of
thing
and
then
they
committed
nuclear
war
on
each
other
they
had
to
restart
it
and
there
was
more
nuclear
war
and
in
the

opposite

when

they

had

the

least

authoritarian

people

in

there

they

weren't

going

I

will

bomb

you

or

you

give

me

all

your

money

type

of

thing

they

were

going

let's

try

and

work

together

even

though

it

was
meant
to
be
a
competitive
game
they
had
the
most
cooperative
game
they'd
ever
seen
even
though
400,000
people
in
this
world
scale
dynamic
thing
still
died
and
it's
an
interesting
look
at
the
reflection
of
how

cognitive
rigidity
because
it's
attached
to
right
authoritarianism
because
the
traits
are
being
conventional
authoritarian
authoritarian
aggression
submission
and
conventionalism
cognitive
fragility
is a
strong
factor
in
how
cooperative
people
are
in
political
dynamics
and
also
how
much
political

dynamics
can
be
reduced
to
chaos
and
yeah
a lot
of
ways
to go
with
that
but
I
mean
your
presentation
and
hopefully
the
forthcoming
works
will
bring
some
of
that
thinking
out
more
and
I
look
forward
to
seeing

thank you
so much
anything
else you
want to
say
or ask
no
I mean
it's
been a
pleasure
thank you
for the
opportunity
I'm sorry
if I didn't
do the
presentation
very well
I kind
of
sprinted
through
quite a
bit
of
things
and
such
but
I
really
appreciated
it
thank
you
yeah

hope
people
enjoyed
it
and
follow
up
with
the
contact
if
they
have
any
additional
comments
they'd
like
to
provide
and
good
luck
with
the
work
see ya
bye
yeah
thankry

thankry

Thank you.